Traffic Accident Analysis & Exposure-Adjusted Risk: 5-Minute Talking Points

Visual Analytics Lab, Review 1 | Saranya Dalal [26MML0007], Divya Pujari [26MML0009], Rohan Yadav [26MML0020] | Guide: Dr Manikandan K (SCOPE)

0:00 - 0:30 Opening: the problem

- Most road-safety work ranks roads by **how many crashes** they have.
- Counts follow traffic. Busy roads look dangerous because they carry more vehicles, mostly with minor shunts.
- Quiet rural roads have few crashes, but those crashes are often fatal, so a count-based list misses them.

  Say: "Our claim: one-eighth the crashes, twelve times the risk. Low frequency masks high severity."

0:30 - 1:15 Method: exposure-adjusted severity risk

- 22,000 crash records in UK STATS19 format on 160 road segments, joined to AADT traffic volumes.
- **Exposure** = AADT x length (km) x 365 / 1,000,000, in million vehicle-km (MVKM).
- **Severity weights** (UK DfT injury-cost ratios): Fatal 50, Serious 10, Slight 1.
- **Risk** = weighted crash sum / exposure. Segments with <3 crashes or <0.2 MVKM are suppressed as noise.
- Median splits on count and risk put every segment in one of **4 quadrants**; Q2 = low count, high risk = 'deadly trap'.

1:15 - 2:00 Core finding: the benchmark pair (Core tab)

- **SEG-0417, M25 interchange**: 47 crashes, but risk only 6.86 per MVKM (risk rank #152).
- **SEG-1902, B4009 Chiltern ridge curve**: 6 crashes, risk 84.4 per MVKM (risk rank #2 of 160).
- 48 segments are deadly traps. 9 of the 10 highest-risk segments are in Q2, and their median count rank is 117 of 160.

  Say: "A count-based priority list would never reach these roads. Exposure adjustment puts them at the top."

2:00 - 2:40 RQ1 - WHEN do crashes happen vs. when are they lethal?

- Volume peaks with commuting: 08:00 is the busiest hour (1,908 crashes), with a second evening peak.
- Lethality runs the other way: 05:00 has the highest fatal share (6.8%).
- Night (20:00-05:59) fatal share is 5.3% against 3.5% by day. The 168-cell day x hour heatmap shows where it concentrates.

2:40 - 3:15 RQ2 - WHERE are the true hotspots? (map tab)

- 3D map of every segment: colour by raw count, by exposure-adjusted risk, or by rank shift.
- Rank shift = count rank minus risk rank. SEG-0142 (rural B-road, 20 crashes) jumps from count rank #151 to risk rank #1.
- The leaderboard below the map lists the segments a count-based ranking pushes down the list.

3:15 - 3:45 RQ3 - Under what conditions?

- Unlit darkness: 37.3% of crashes are fatal or serious, against 24.1% in daylight or lit darkness.
- Motorcycles at 70 mph: 54% severe across 1,689 crashes (network average 24.8%).
- Surface alone matters less: dry and wet roads are both about 25% severe. Lighting and speed drive severity.

3:45 - 4:15 ML classifier and what-if simulator

- Random Forest severity classifier. Top drivers: vulnerable road user (0.37), AADT (0.15), speed limit (0.12).
- Simulator demo: 60 mph + vulnerable user = 50%. Add dark night and no street lighting and it rises to 90%.
- Each score comes with an engineering recommendation: LED lighting, speed cameras, high-friction surfacing.

4:15 - 4:40 India transferability case (Bengaluru)

- The **Study Context** switch re-points the whole dashboard to 5 Bengaluru corridors (2,500 crashes).
- Two-wheelers and pedestrians are 58% of crashes. Night fatal share is 7.1% against 4.9% by day.
- Sarjapur Road has 0.97x the crashes of Outer Ring Road but 4.8x the risk. Kanakapura NH-948 ranks highest (56.8 per MVKM).

  Say: "The method transfers: the same inversion appears in mixed Indian traffic."

4:40 - 5:00 Pipeline and close

- Python ingest -> preprocessing -> DuckDB exposure and risk -> pre-aggregated Parquet cubes. The full run takes about 5 seconds.
- Two front ends read the same cubes: the Streamlit dashboard and a 7-page Power BI report.

  Say: "Close: rank roads by risk per vehicle-km, not by crash count. Otherwise the deadliest roads stay hidden."

If asked	Short answer
Is the data real?	The records are generated by pipeline stage 1 in the STATS19 schema. A real STATS19 extract can replace them without code changes.
Why 50 / 10 / 1?	They follow UK DfT injury-cost ratios: a fatality costs far more than a slight injury.
Why suppress segments?	Very small counts or exposure make risk swing wildly, so those segments are flagged instead of ranked.
Why DuckDB?	It is embedded and columnar, needs no server and reads Parquet directly.